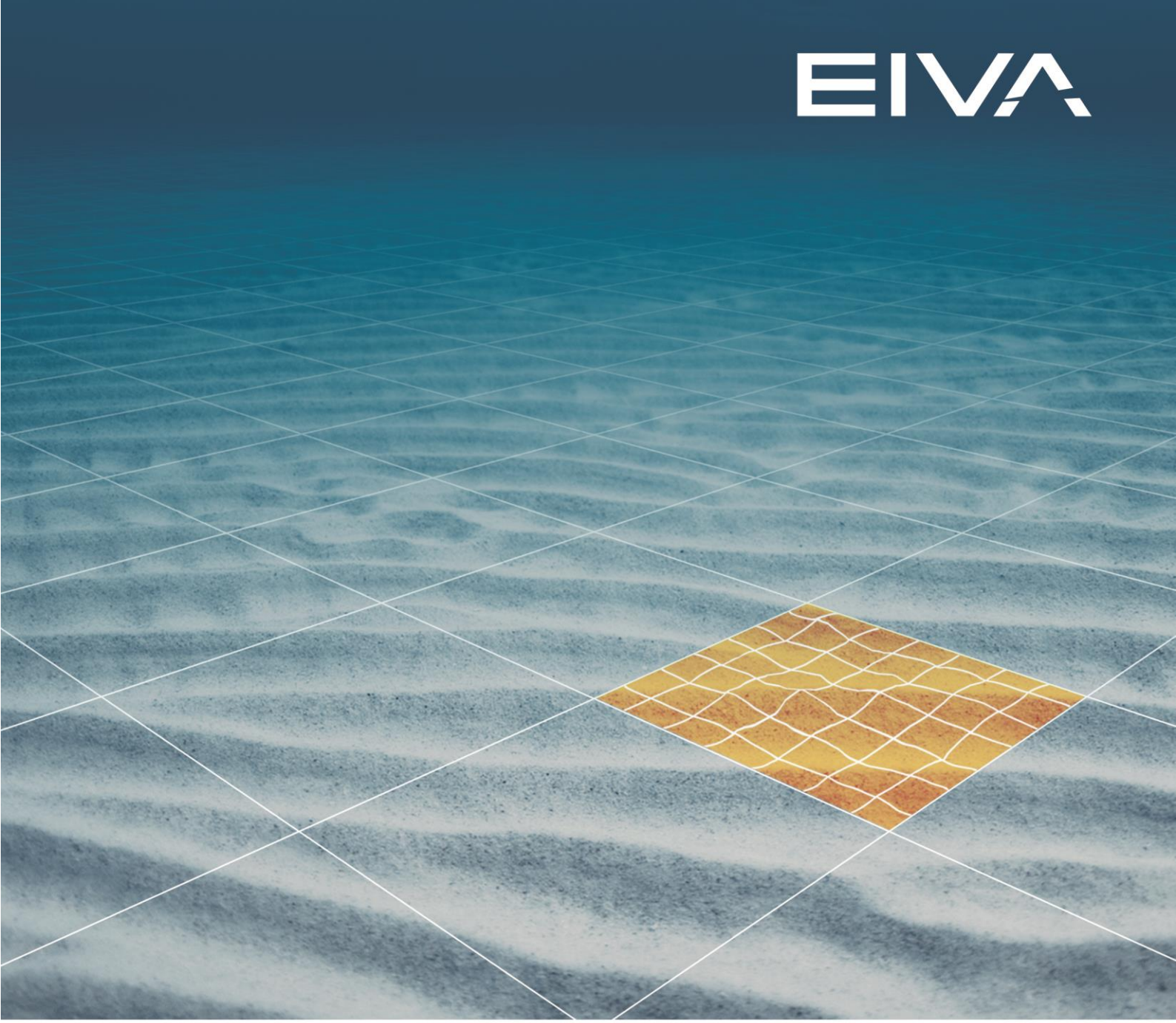

NAVISCAN

NAVISCAN CONFIGURATION

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1 Introduction

The purpose of **NaviScan** is to collect data from multibeam sensors and to generate a file with exact positions of each scan. To do this, navigation information is needed along with gyro, roll/pitch/heave and bathy sensor.

Due to time tagging considerations the **NaviScan software package** is split into two parts:

1. **NaviScan Config** (the Off-line/Setup part) with several utilities for setting up the system and handling the raw data files.

   This guide covers this part only.

2. **NaviScan Online** (the Online/Acquisition part) taking care of the data acquisition and data logging to raw data file/s.

In **NaviScan Config** the user can specify and configure external interfaces, including geodesy settings, interface identification, sensor data format, sensor C-O and exact location on the vehicle (eg vessel/ROV) of each sensor.
All information is written to the configuration file **NaviScan.BIN**, which is read by the **NaviScan Online** program before the data acquisition starts.

1.1 Related manuals

The guide to NaviScan Online can be found in NaviScan Configuration menu bar under Help / NaviScan General Course.
A sensor list with their data formats are described here **Help / Data Formats** (*NaviScan reference manual*).

2 Getting started

After installing NaviScan, two desktop shortcuts to NaviScan Config and NaviScan Online are created. Start the **NaviScan Config** program from the desktop shortcut or activate it from the **Windows Start menu**:

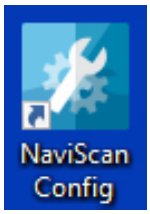


Figure 1 Desktop shortcut for NaviScan Config

On first start the message box opens asking the user if he/she wants to make a new setup. Press **Yes**.

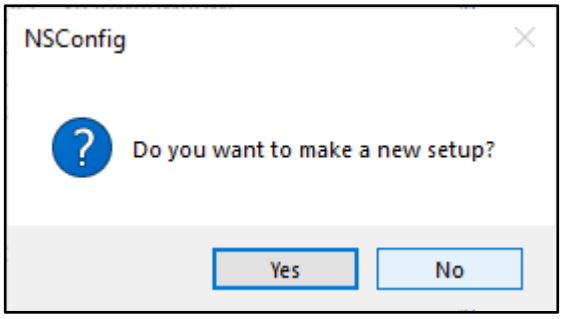


Figure 2 NaviScan Config: Make a new setup

NaviScan opens with the default layout. The following screenshot shows the UI elements.

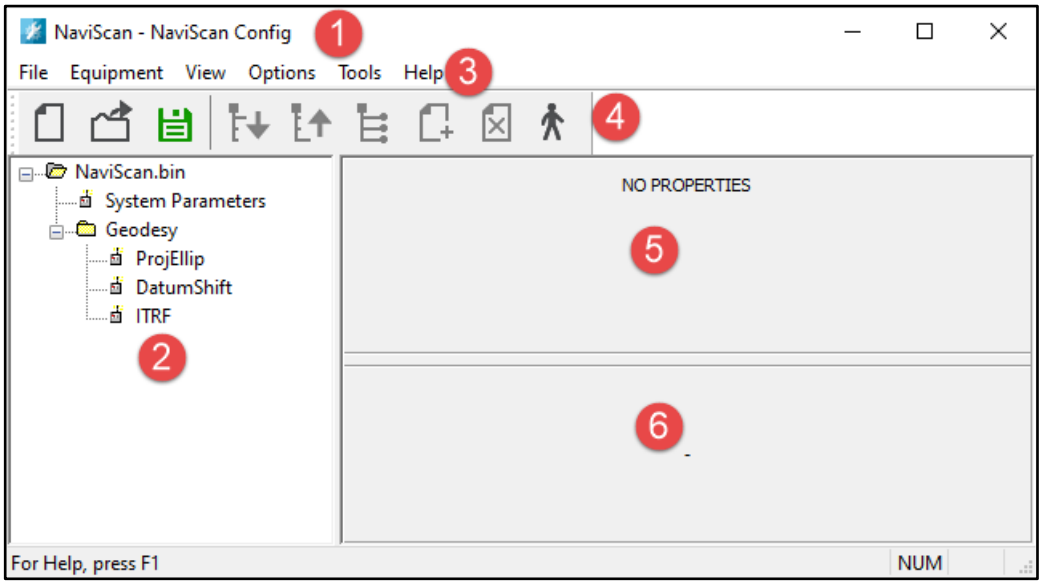


Figure 3 NaviScan Config layout: (1) Configuration file information bar, (2) Tree view or Explorer panel, (3) Menu bar, (4) Toolbar, (5) Property view or Information window, (6) Help and status bar

2.1 The configuration file NaviScan.BIN

In the NaviScan.BIN file all data is stored.
It must not be renamed and stored to the C:\EIVA\NaviScan\Db\. folder.

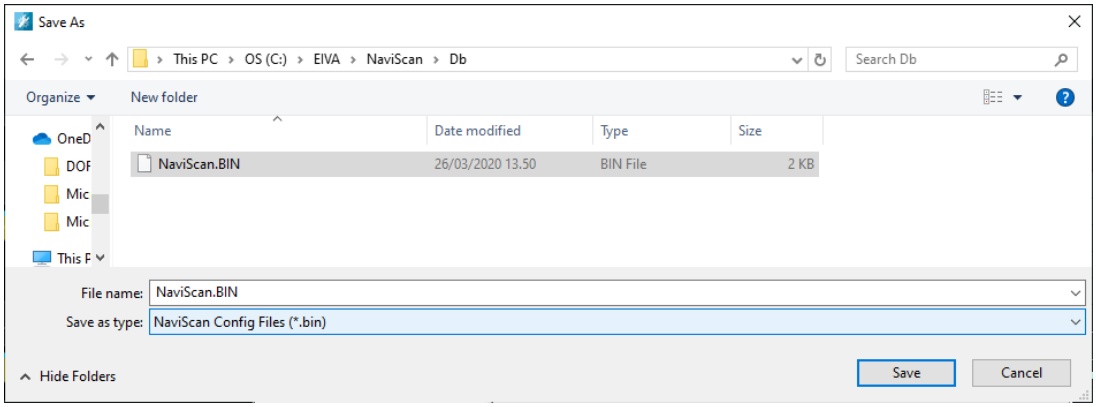


Figure 4 Save as file type NaviScan Config Files (*.bin)

Note that the file must be a binary file in **NaviScan Config** format in order to make any sense to the **NaviScan Config** program. Therefor use the option Save as type **NaviScan Config Files (*.bin)** as shown in the screenshot above.

This file is also used by the NaviScan Online program. You can use two or more configuration files but note that only the file named **NaviScan.BIN** is used by the online program.

2.2 Tree view or Explorer panel

The **Tree view** has three different levels:

- **File level**
  Here the name of the configuration file is visualized at the very top of the Panel
- **Sensor type or Instrument type level**
  Here the instrument types are listed. An instrument type could for instance be **Echosounder**
- **Individual sensor or Instrument level**
  Here instruments are listed. Instruments are organized relative to their type. In the list to the left, Instrument Type **Gyro** has two instruments in the child level, **NMEA and Ship's Gyro**

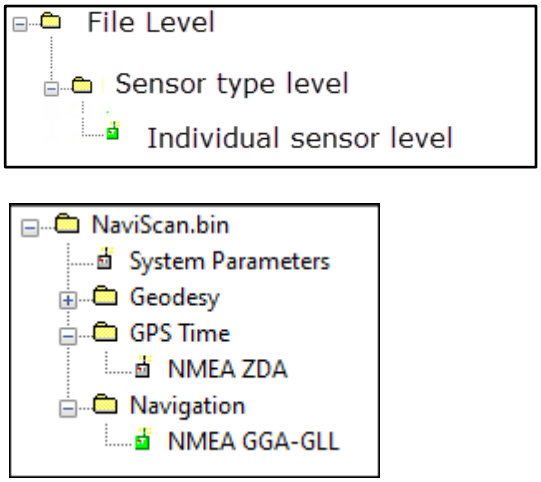


Figure 5 An example of the Tree view with levels

2.2.1 File level

This represents the top level of the configuration. Right-click on an item on the file level gives you the option to, eg **Expand tree.**

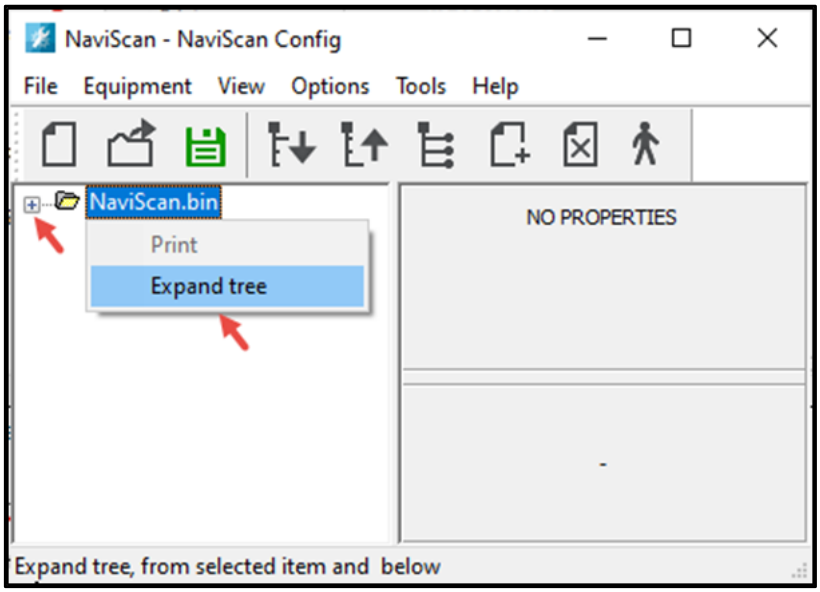
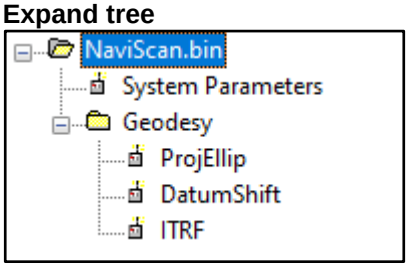


Figure 6 Context menu of NaviScan.BIN

Here are all options on **File level** listed:

- **NaviScan.bin**
  - **Expand tree**



Expands or collapses the tree.
Shows the default NaviScan configuration including System Parameters and geodesy settings.

Shows overview of sensor configuration

NaviPac	UDP 5000 127.0.0.1
Octans3000 Gyro HQ	ATTU 5002 192.168.1.11
Octans 3000 Motion	ATTU 5002 192.168.1.11 XYZ 0.000 0.000 0.000
ProFlex500 Bathy	ATTU 5001 192.168.1.11 XYZ 1.379 1.487 10.015
Ashtech Proflex500	ATTU 5001 192.168.1.11 XYZ 1.379 1.487 10.015
Reson7k RawDetect7027	TCP 7000 127.0.0.1 XYZ 1.840 -0.235 -1.589
Reson Snippet7028	TCP 7000 127.0.0.1 XYZ 1.840 -0.235 -1.589

- **System Parameters**

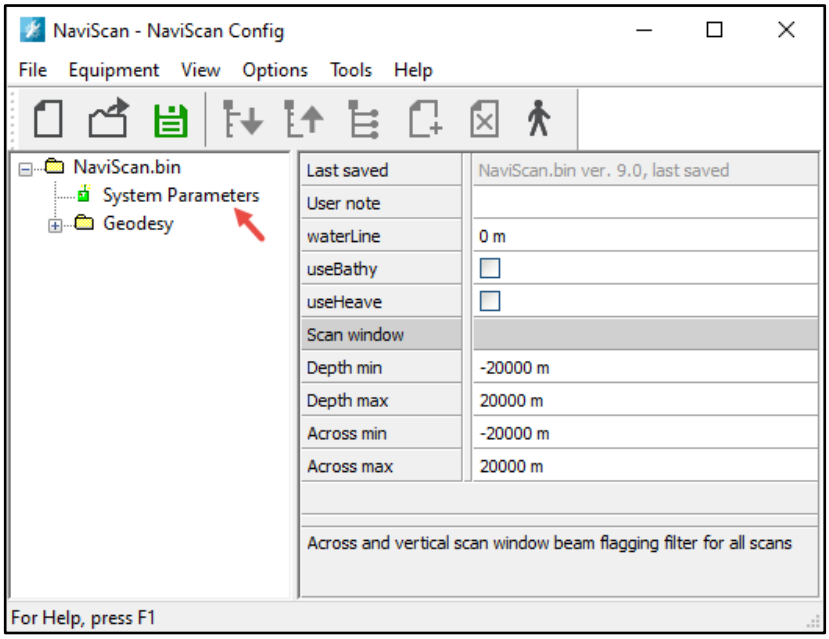


Figure 7 Properties of the System Parameters

- **User note**
  It gives name and date of the current setup and it allows the operator to type relevant information, eg who-am-I etc. It's recommended that this information is filled, as it's stored together with the configuration data.
- **waterline**
  Offset from reference point to water line, positive up
- **useBathy**
  Enable use of bathy in result calculation (else waterline relative)
- **useHeave**
  Enable use of heave in result calculations (normally not for ROV)
  Specifies whether the heave from RPH sensor should be used.
  When mounted on a ROV the heave must be disabled and a bathy sensor must be defined.

Note: The heave will be recorded anyhow so you can reselect this in NaviEdit.

- **Scan Window**
  Across and vertical scan window beam flagging filter over all scans
- **Depth min**
  System z filter min window in absolute reference system with bathy included, -20000 to 20000 meter positive down
- **Depth max**
- **Across min**
  System x filter min window rel ref point, in local vessel reference system, -20000 to 20000 meter positive starboard
- **Across max**

- **Geodesy**

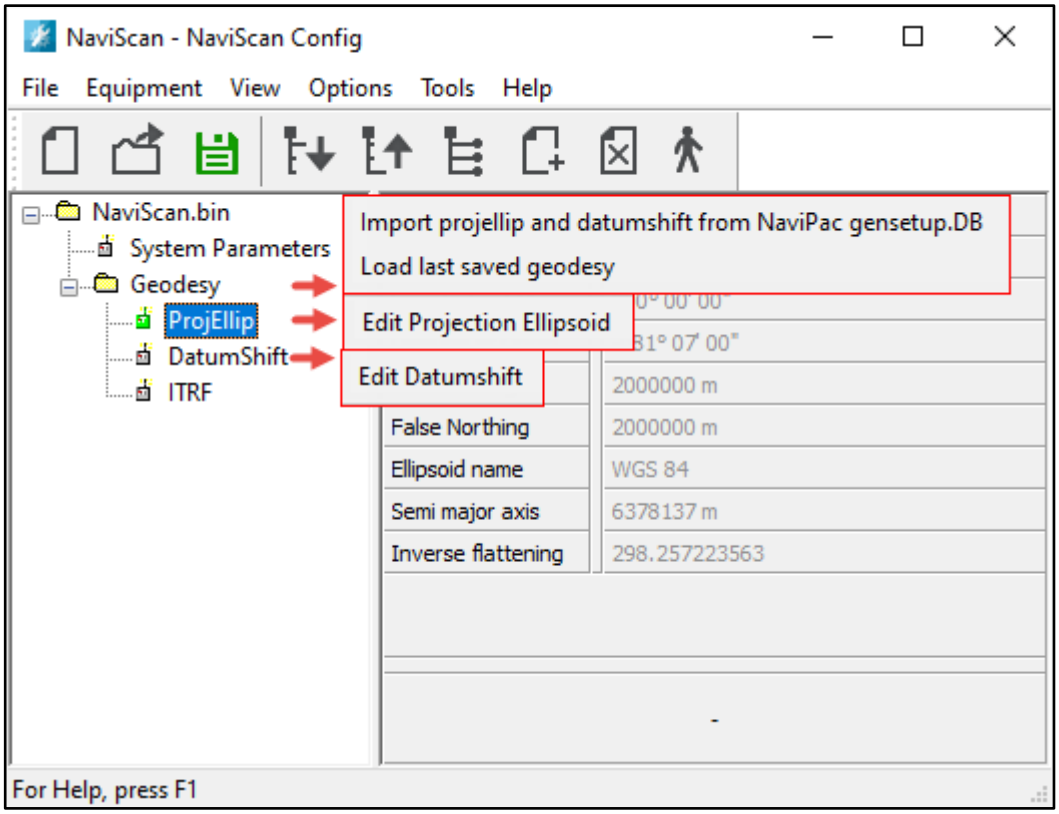


Figure 8 Geodesy context menu's

- **Import projellip and datumshift from NaviPac gensetup.DB**
  Allows to import a gbin file from NaviPac.
- **Load last saved geodesy**

  ProjEllip

- **Edit Projection Ellipsoid**
  Allows to change the geodesy settings
- **DatumShift**
  Edit Datumshift
- **ITRF**

2.2.2 Sensor type level

On a sensor type that support multiple sensors of same type, you can use the option **Add New** to add a new sensor. It will lead you through all steps of adding the sensor, include eg Sensor type, port. Offset needs to be set in the sensors property view.

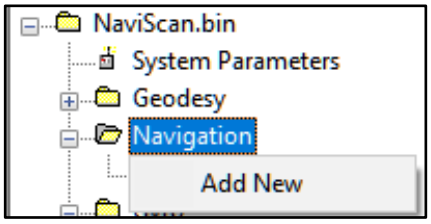


Figure 9 Context menu Add new on Sensor type level

In the following it is shown how to add a new sensor.
Choose the sensor type you want to add, eg here from the sensor type GPS Time. Search by typing its name in the **Search Filter** to quickly find the sensor.

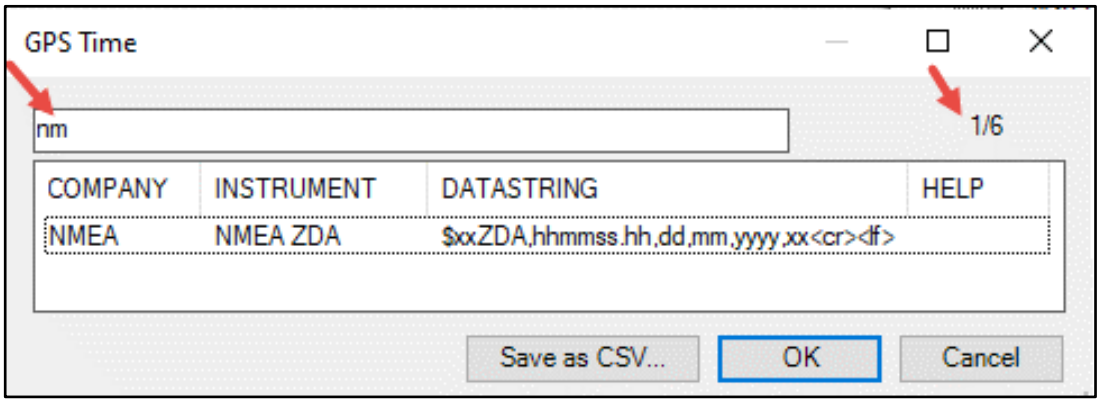


Figure 10 Instrument search option, Company, Instrument, Datastring and Help. Number of predefined sensors.

After selecting the sensor, the **Port settings** view opens, where the port is configured.

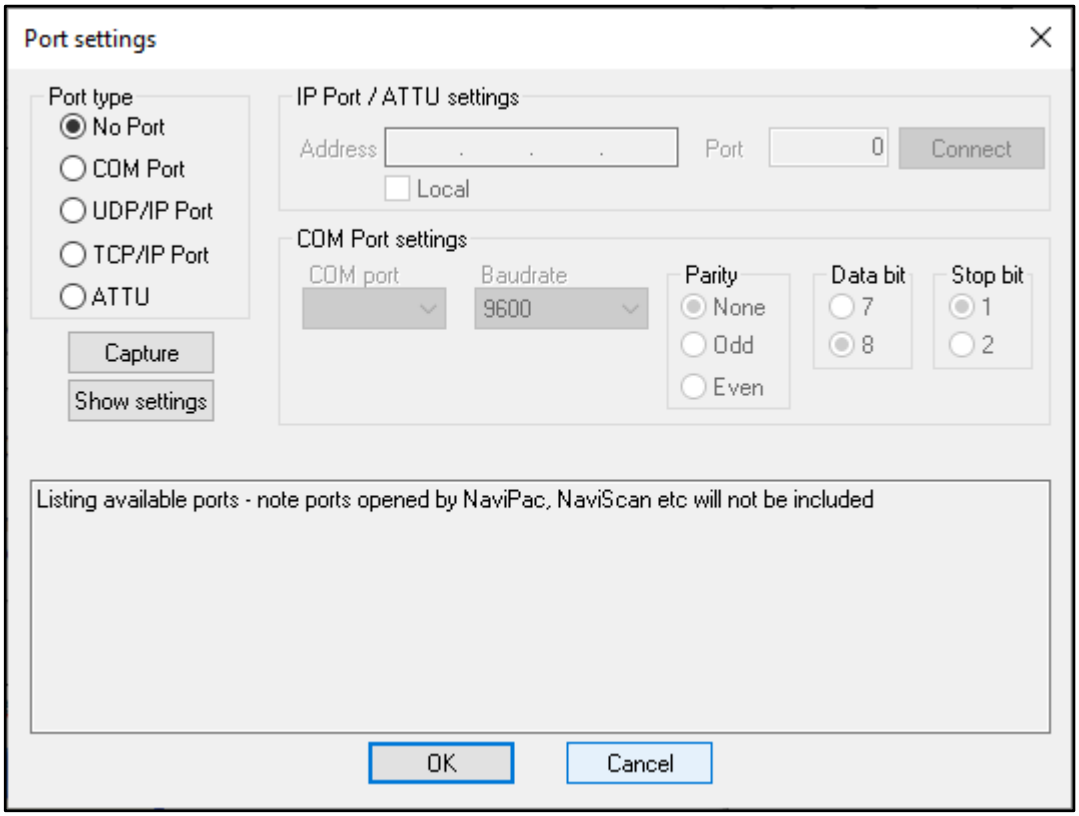


Figure 11 Port settings eg serial communication for the used sensors.

After port settings, NaviScan opens and in the sensor property view, offsets can be entered.

2.2.3 Individual sensor level

This represents the actual sensors.
If more sensors of a kind are used, use the **Move up/Move down** option to set the order.
The uppermost sensor is used as the **primary** and is the one used in NaviScan Online. The rest is logged and can be used at a later stage in eg NaviEdit.

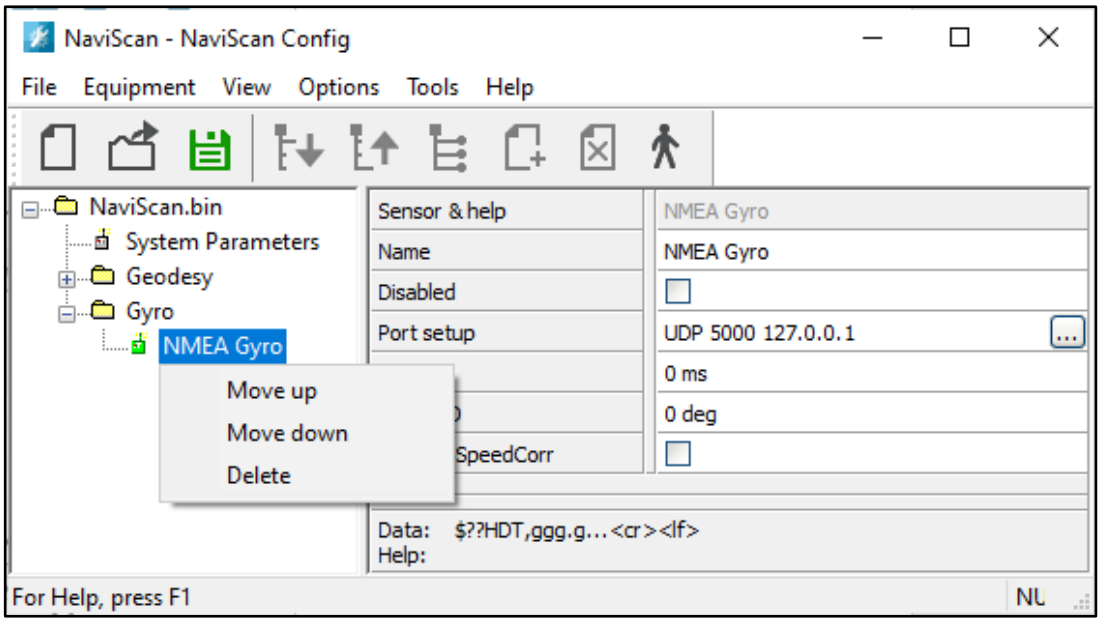


Figure 12 Context menu on sensor level, Move up, Move down, Delete

On sensor types, it is possible to add by right clicking on the type in the tree window.

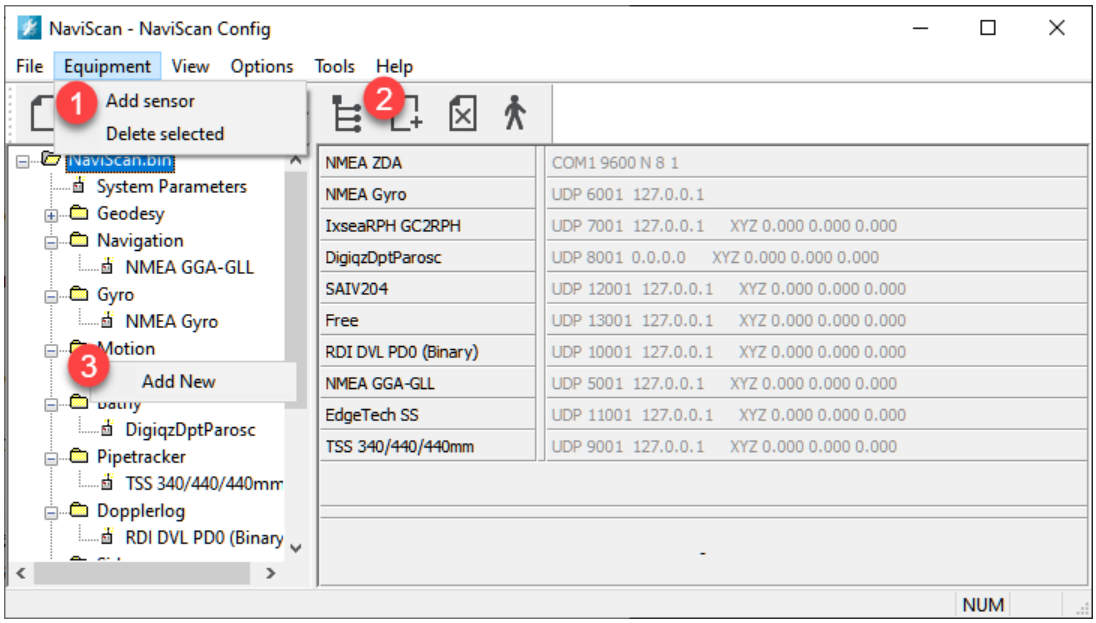


Figure 13 NaviScan: Three options to add a sensor

3 The Toolbar

The toolbar has the following items (from left to right):

- **New**
- **Open**
- **Save**
- **Move sensor down/Move sensor up**

  Move the sensor up and down in priority. The sensor that in the top is the **primary**, and is the one, used in NaviScan Online. The rest is just logged and can be used at a later stage in eg NaviEdit.
- **Expand tree**

- **Delete**

  Delete the sensor from the configuration. This can also be achieved by pressing **Delete** on the keyboard.
- **Start NaviScan Online**

The toolbar can be disabled (from the View menu) and it can be moved to another location. The toolbar functionalities are also accessible from the main menu of the NaviScan Configuration program or from the Explorer Panel.

4 Menu bar

The menu bar consists of six options: **File, Equipment, View, Options, Tools, Help**. These are described in the following:

4.1 File

The **File** menu has options used to handle the NaviScan configuration file.

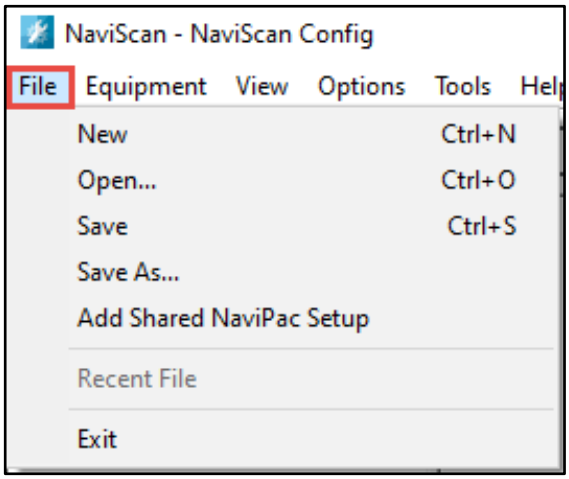


Figure 14 NaviScan Config: File options

- **New**
  Standard Windows facilities for file handling. Here the NaviScan.BIN file can be created, saved and opened. The name of the configuration file, to be used online, must always be NaviScan.BIN
- **Open…**
  Opens an existing configuration file for editing. By selecting this item a standard Windows file dialogue is opened to select filename, directory and drive.
  The NaviScan.BIN should be stored at this location, C:\EIVA\NaviScan\Db\.
- **Save**
  Saves the configuration to the file opened. The filename, drive and directory are shown at the top of the **NaviScan Configuration** screen.
  The NaviScan.BIN should be stored at this location, C:\EIVA\NaviScan\Db\.
- **Save As…**
  Saves the configuration file using a standard file dialogue to specify filename, directory, drive for the file to be saved. The **Save As** command is useful when using two or more configuration, but note that only the file named **NaviScan.BIN** is used by the NaviScan Online program.
- **Add Shared NaviPac Setup**
  todo
- **Recent File**
  If you maintain more files, this shows the last four files.
- **Exit**
  Terminates the program with prompting for saving current settings.

4.2 Equipment

Choose the Equipment entry to add a new sensor/instrument, among a list of types and drivers, to the configuration. Note that the items of the menu are greyed and cannot be selected, unless the Explorer Panel has focus (move the mouse cursor to the Explorer Panel).

NaviScan Config has a large set of predefined sensors or the user can specify a data input format him/herself.
The data format of the predefined sensors is specified in chapter 1.1.

Choose the **Equipment** option and choose **Add sensor**. Now a list of available **sensor types** is shown:

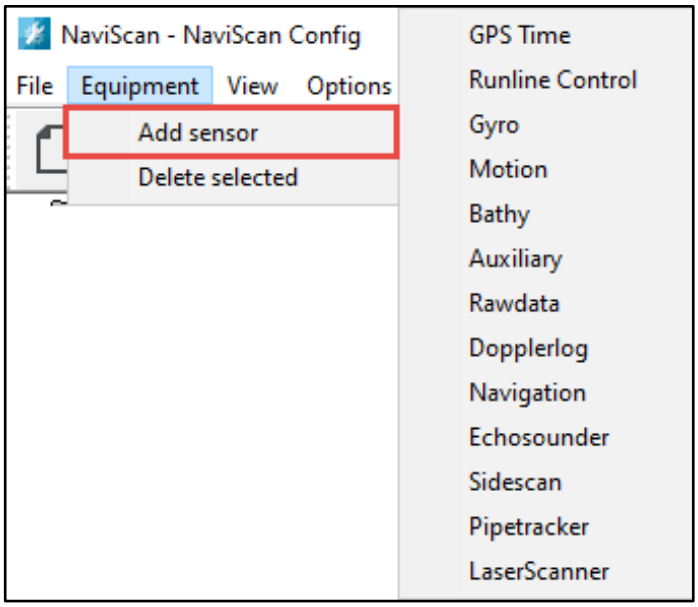


Figure 15 Equipment dropdown menu with sensor types

4.2.1 GPS TIME

Select a sensor type and proceed:

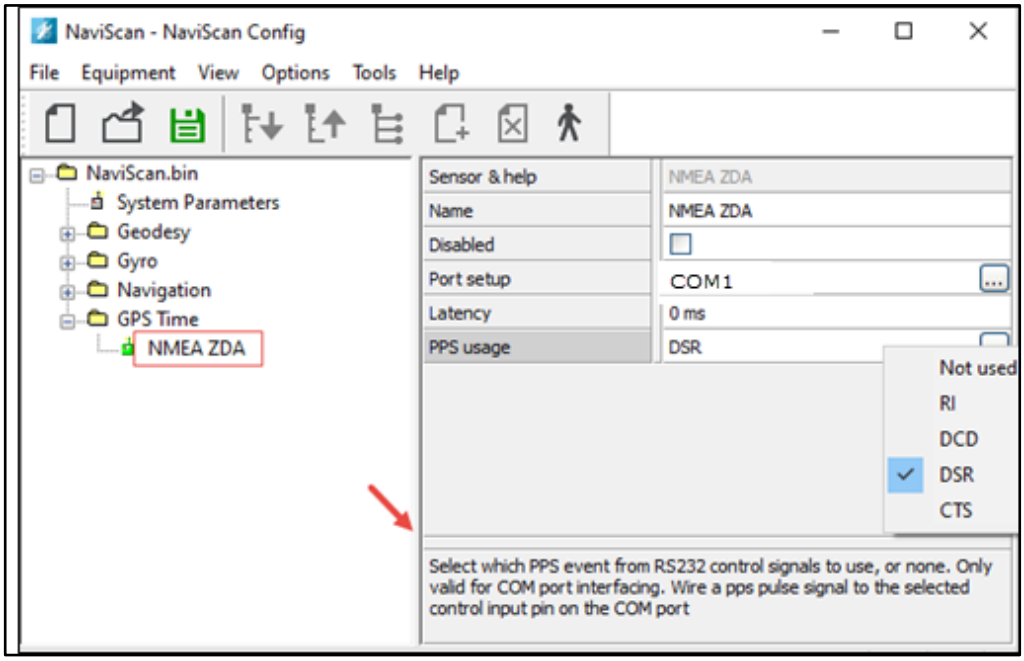


Figure 16 NMEA ZDA on COM port 1

PPS usage: Select which PPS event from RS232 control signals to use, or none.

- **Sensor & help**
  Expected data format is shown, here $xxZDA,hhmmss.hh,dd,mm,yyyy,xx<cr><lf>
- **Disabled**
  Disable use of the sensor so that it will not be used online, and will not produce alarms and will be placed at bootm of sensor list when setup is saved.
- **Port setup**
  Always enter ip addr of the source sending data, or localhost if sent from local pc. For UDP/ATTU the ip addr is used as filter, only passing data from that ip, you can use ip addr 0.0.0.0 if you want to disable the UDP filter and get data from all ip that sends to the port. Note: that the port setup dialog test Connect button for UDP, in contrast to naviscan online, receive and display data from all ip addr sending to the port, so you cannot be sure that athe incoming test is from entered ip.
- **Latency**
  Latency corrections in ms, that will be subtracted from the timestamp when receiving each packet in order to correct for timestamp to be valid when data is measured, normally set to 0 when data carries its timestamp in the packet, but might be needed on COM port interfaced sensors, as some motion sensors have aknown delay of eg 25 msec.
- **PPS usage**
  Select which PPS event from RS232 control signals to use, or none. Only valid for COM port interfacing. Wire a pps pulse signal to the selected control input pin on the COM port.
  You may use the time synchronisation without PPS, NaviScan will then use the timestamp in the ZDA/UTC message and consider it to be real-time. This is most

often not accurate enough for MBE acquisition!

It is normally important that the NaviScan computer time is synchronised with the sonar time.

4.2.2 RUNLINE CONTROL

This is data communication with NaviPac. Runline information is retrieved every second.

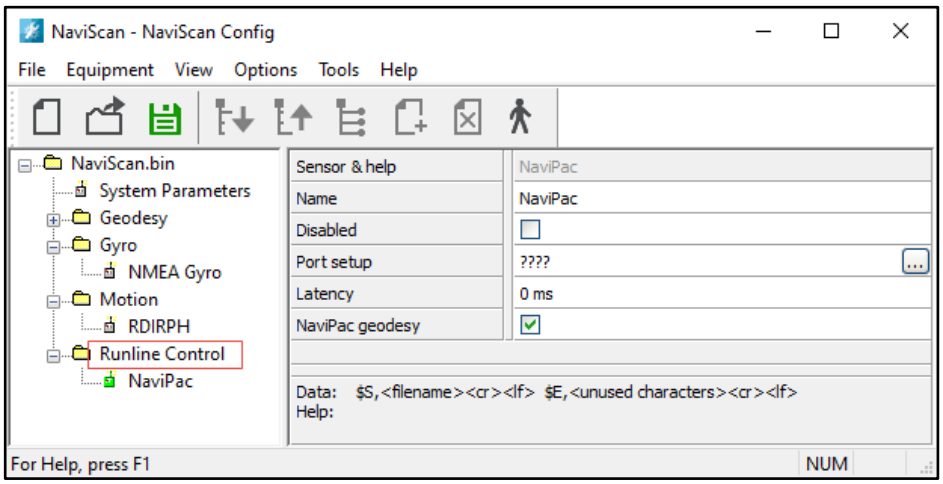


Figure 17 Runline Control

- **Sensor & help**
  Expected data format
- **Name**
- **Disabled**
- **Port setup**
- **Latency**
- **NaviPac geodesy**
  Use geodesy packets received from NaviPac as logged and applied geodesy, this will overrule the nsconfig geodesy. Until such geodesy packets are received, the nsconfig geodesy will be used as backup.

4.2.3 GYRO

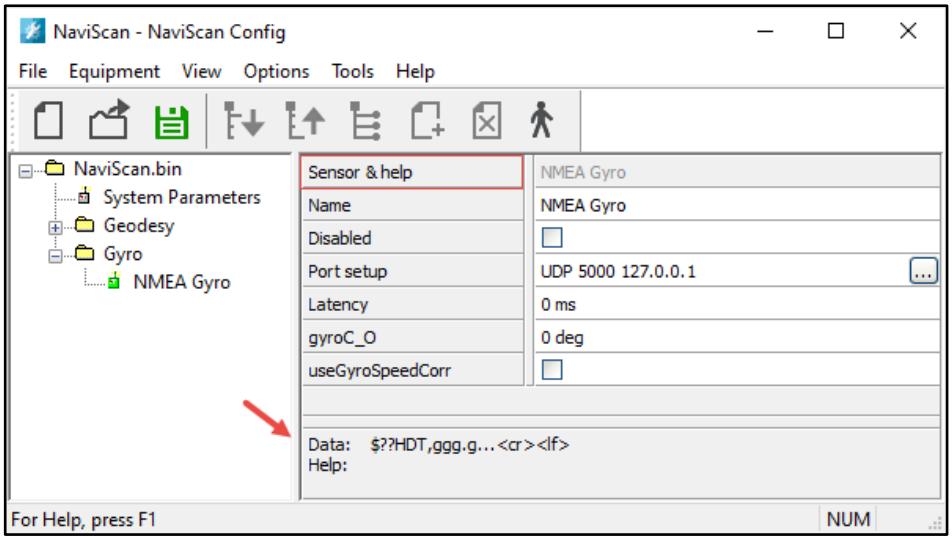


Figure 18 Sensor NMEA Gyro with Sensor & help option showing the expected data format

At gyro setup the gyro system is identified; the gyro offset must only be defined when free format is selected:

- **Sensor & help**
  Expected data format
- **gyroC_0**
  C-O value for this sensor
- **useGyroSpeedCorr**
  Use dynamic gyro correction

4.2.4 MOTION

When configuring a motion sensor, the interface for the roll, pitch and heave sensor (RPH sensor) is defined:

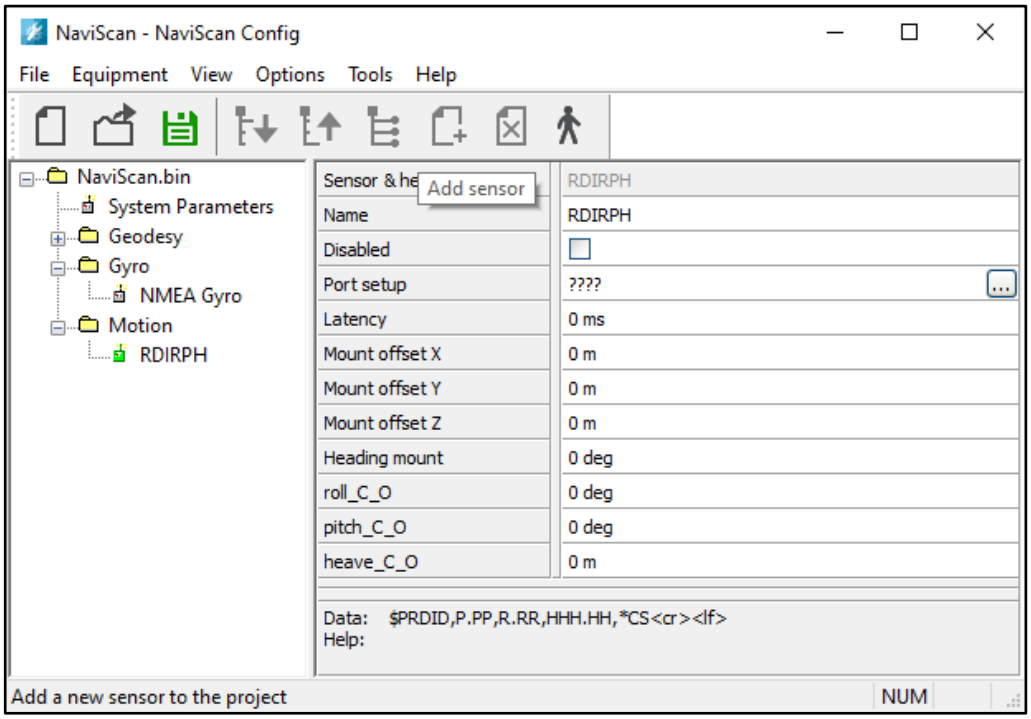


Figure 19 Interfacing Motion sensor

- **Sensor & help**
  String layout
  Specifies the offsets to the RPH information in the serial string by defining the number of characters before the field.
  Roll Roll value "rrr.r" in degrees
  Pitch Pitch value "ppp.p" in degrees
  Heave Heave value "hhhh" in cm
  Termination character, the ASCII number of the last character – eg. 10 for <LF>
- **Heading mount**
  Mount heading angle relative to reference frame, positive clockwise
- **Roll_C_O**
- **Pitch_C_O**
- **Heave C O**

4.2.5 BATHY

If mounted on a ROV, bathy sensor must be defined to give the exact depth of the ROV, otherwise go to the **System Parameters** and disable the **useBathy** option.

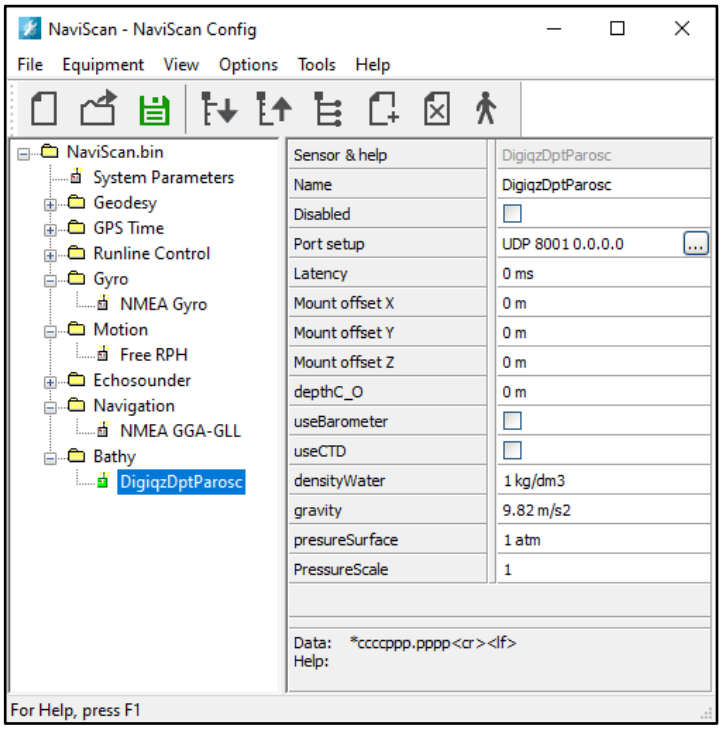


Figure 20 Bathy properties

4.2.5.1 String Layout

Specifies how NaviScan shall read the data from the string

- Bathy depth
  Expected data format
- Termination character
  The ASCII number of the last character – eg 10 for <LF>

4.2.5.2 Pressure -> Depth calculation

For the sensors giving pressure instead of depth (eg Digiquartz), you must specify the following three parameters:

- Water density
- Gravity
- Pressure at surface
- Pressure Scale

Water Density, Pressure of surface and Gravity can be modified by user, and those settings will be used in the depth calculation by applying a simple Saunders and Fofnoff:
*Depth = (Pressure – PressureAtSurface)/(1000 * Gravity * WaterDensity)*

Note: For the **HoneywellPPTR20** sensor set Pressure at surface to 0, because this sensor internally is connected to the surface pressure, and therefore subtracts itself the surface pressure (outputs the differential pressure).

4.2.5.3 Vessel mounted

Please note that the bathy driver also utilises a series of other related inputs:

- **NaviPac Heave correction**
  For vessel mounted solution this input gives the GPS based height of the vessel CRP, from NaviPac.
- **GPSHeightGGA**
  For vessel mounted solution this input gives the GPS based height of the vessel GPS antenna
- Altimeter
  Various drivers with ROV altimeter input – is just treated as secondary input compared to the MBE
- **Barometer**
  Input of surface pressure [in hPa] – will be subtracted from ROV depth

4.2.6 AUXILIARY

NaviScan offers special auxiliary input for sensors that doesn't match the other sensors

COMPANY	INSTRUMENT	DATASTRING	HELP
Franatec	Methane	M<ppmv>, R<ppmv>, T<degC>, H<transmission %>, L<humidity %> <cr><lf>	
Reson	Reson (Seabird)	T<temp>, C<cond>, P<pres>, S<sal>, c<svel> <cr><lf>	
Saab	Pressure Saab	... *0001<psi><cr><lf>	
Saab	SVS Saab	... <m/s><cr><lf>	
SAIV	SAIV204	N<id> C<cond> T<temp> P<pres> c<svel> S<sal>	
Valeport	VpBathyPack	dd/mm/yyyyhh:mm:ss<cond><pres><temp><alti>[...]<cr><lf>	

Figure 21 Reson, SAIV, Valeport

Auxiliary data will be stored as raw packets in NaviScan without any display or control, but will be available in NaviEdit for further use.

4.2.7 RAWDATA

NaviScan offers a raw data input

COMPANY	INSTRUMENT	DATASTRING	HELP
	Free		no help

Figure 22 Sensor interfacing: Raw data

A raw input will read into the online program and recorded in the SBD format. There will not be any data display or handling in NaviScan or NaviEdit.

4.2.8 DOPPLERLOG

If NaviScan is operated in areas where the raw navigation is outside the specifications for precision, you may improve it by using a Doppler Velocity Log. By combining position, gyro and velocity, you get the best overall result.

COMPANY	INSTRUMENT	DATASTRING	HELP
EDO	EDO 3050 287.5 kHz (low)	EDO 3050 287.5 kHz (low)	
EDO	EDO 3050 596.5 kHz (med)	EDO 3050 596.5 kHz (med)	
EDO	EDO 3050 894.5kHz (high)	EDO 3050 894.5kHz (high)	
Eiva	Eiva Interpreted	Eiva Interpreted	
Nortek	Nortek DF21-DF22	DF21 bottom track DF22 water track	UDP listening
Teledyne RDI	RDI DVL PD6 (Ascii)	RDI DVL PD6 (Ascii)	
Teledyne RDI	RDI DVL PD0 (Binary)	RDI DVL PD0 (Binary)	
Teledyne RDI	RDI DVL PD4 (Binary)	RDI DVL PD4 (Binary)	
Teledyne RDI	RDI DVL PD3 (Binary)	RDI DVL PD3 (Binary)	
Ulvertech	Ulvertech D.A.T.S	Ulvertech D.A.T.S	

Figure 23 Sensor interfacing: EDO, Teledyne RDI, Ulvertech

Data from Dopplerlog is assumed to be in real-time, eg no time delay can be specified.

Sensor & help	RDI DVL PD6 (Ascii)
Name	RDI DVL PD6 (Ascii)
Disabled	☐
Port setup	????
Latency	0 ms
Mount offset X	0 m
Mount offset Y	0 m
Mount offset Z	0 m
Roll mount	0 deg
Pitch mount	0 deg
Heading mount	0 deg
Exclude water tracking	☐
Exclude bottom tracking	☐

Data: RDI DVL PD6 (Ascii)
Help:

Figure 24 DopplerLog: RDI DVL PD6 (Ascii)

4.2.9 NAVIGATION

In this dialogue, you can select the positioning system supplying the navigation information:

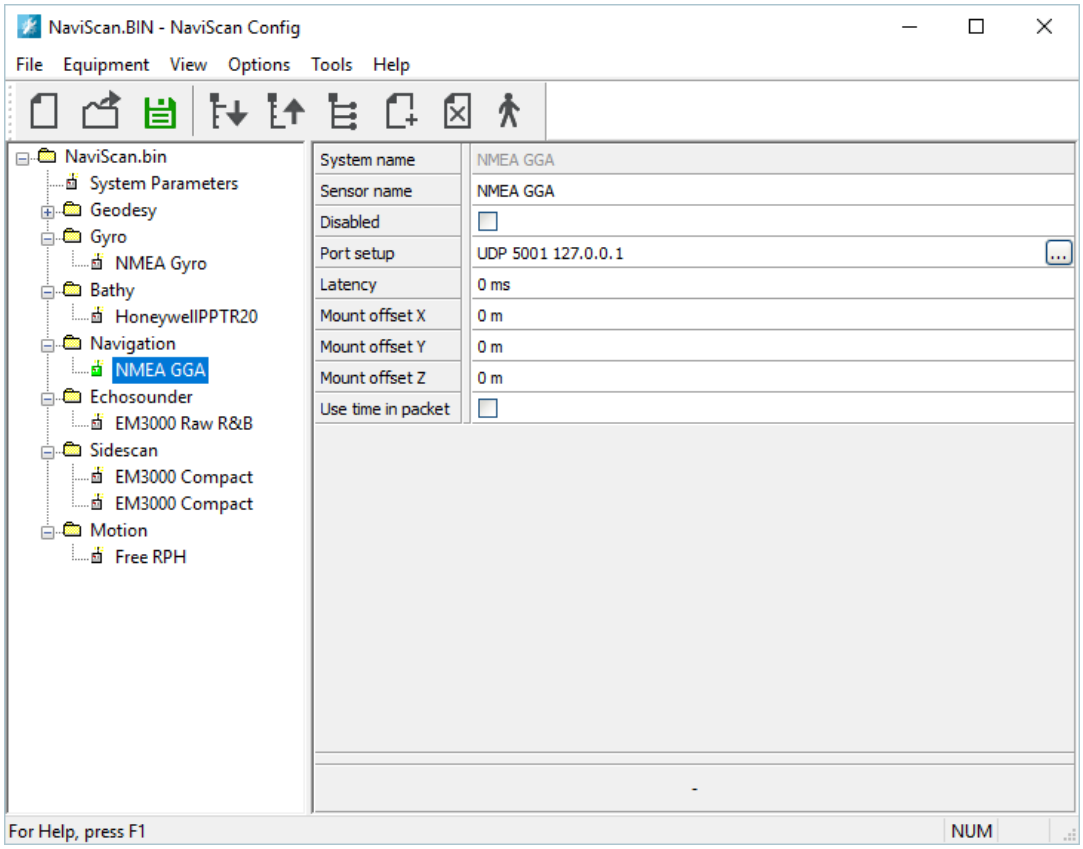


Figure 25 Positioning setup

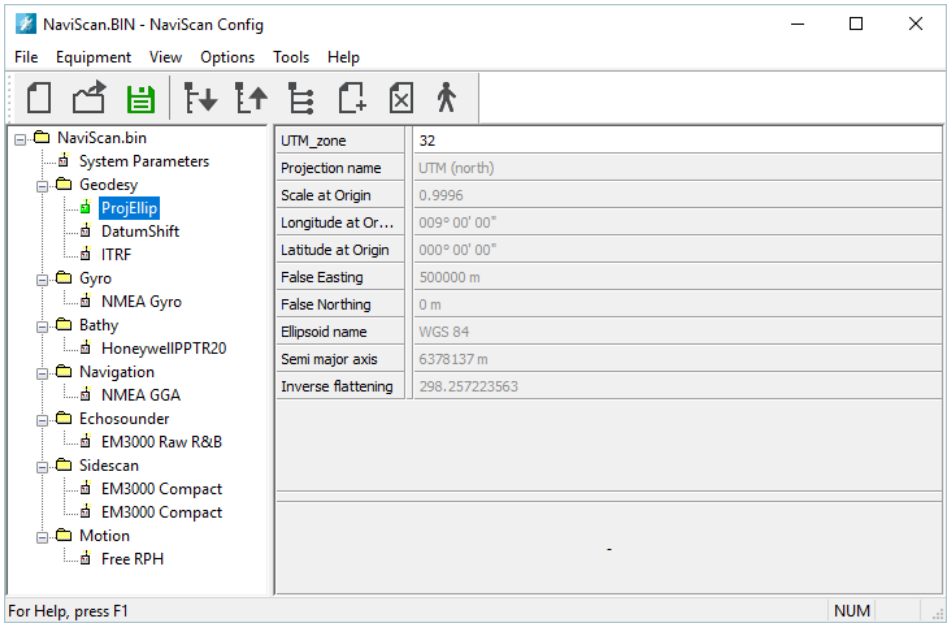


Figure 26 Geodesy settings

For the positioning equipment following data has to be defined when free format is selected:

4.2.9.1.1 Age of data

This defines the age of the data from navigation system. The age can either be a constant delay or specified in the serial string.

- **Fixed**
  Defines a constant delay in milliseconds. Corresponds to the Extra latency.
- **Offset**
  The string includes the age given as milliseconds and this field points out where NaviScan must read the field by defining the number of chars before it.
- **Absolute**
  The string includes an absolute timestamp which is used to determine the latency.
- **Extra latency**
  Here you may add an extra age of data

4.2.9.1.2 String layout

This defines (for non-fixed systems) where NaviScan reads the various data fields by defining the number of characters before the field:

- Time (Optional) Where to find the timestamp in the telegram (format "hh:mm:ss")
- Easting Where to find easting co-ordinate in the telegram (format "EEEEEEEE.EE")
- Northing Where to find northing co-ordinate in the telegram (format "NNNNNNNN.NN")

4.2.9.1.3 Projection, Ellipsoid and Datum shift

If you are using a position input with geographical coordinates – eg NMEA GGA or Applanix POSMV, then you need to tell NaviScan how to convert these to easting and northing. This is done via standard definitions similar to NaviPac. Please refer to NaviPac manual for further information (https://eiva.freshdesk.com/support/solutions/articles/43000520809-geodetic-control).

4.2.10 ECHOSOUNDER

NaviScan Config predefined echo sounders are 2GRobotics, Atlas, Benthos, BlueView, CathX, EdgeTech, Hi-Target, Imagenex, ITER Systems, Klein, Kongsberg, L-3, NewtonLabs, Norbit, Picotech, R2Sonic, Reson, Sonardyne, Teledyne Odom, Tritech, Wassp.

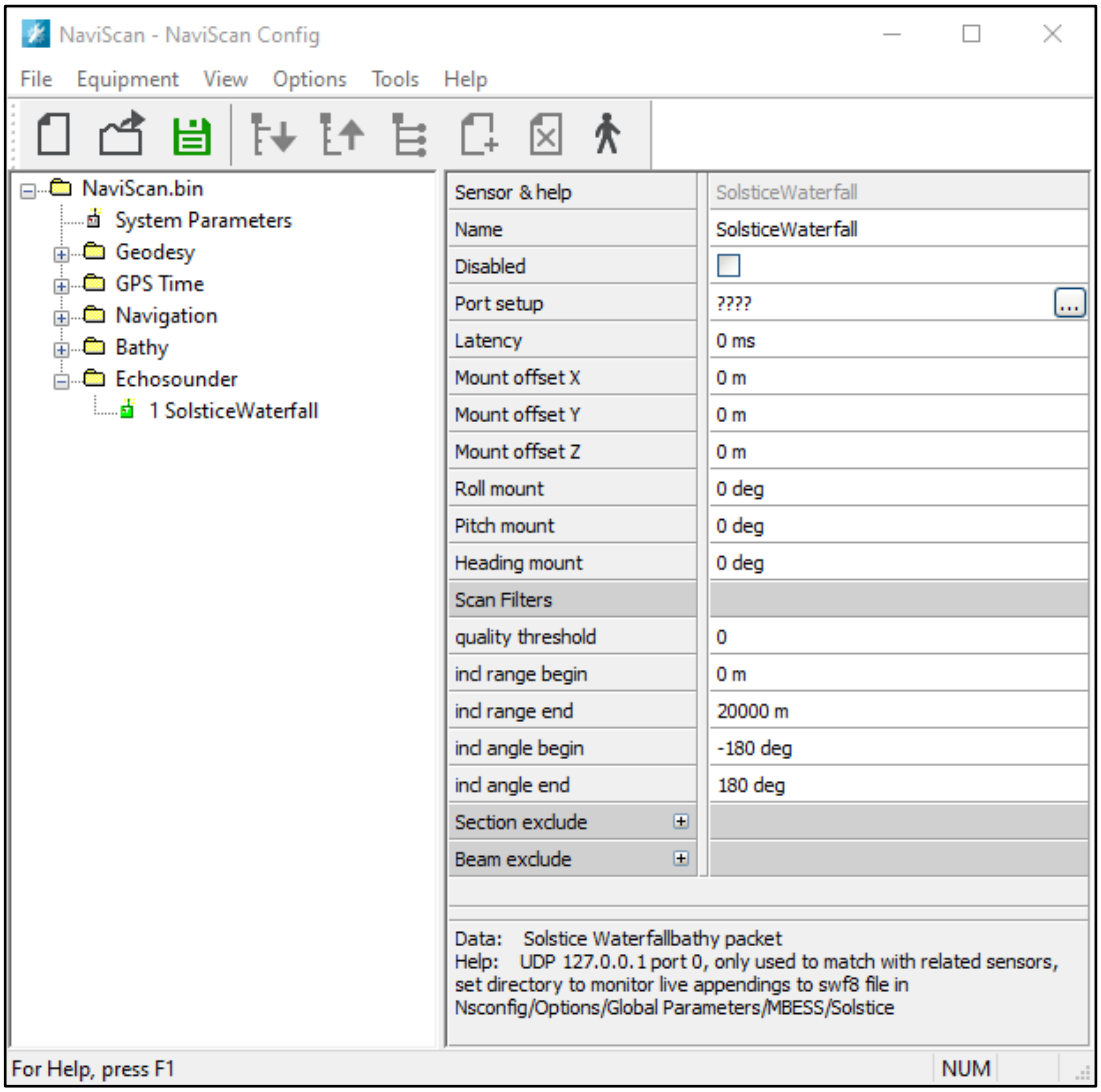


Figure 27 An example of echo sounder interfacing, and sensor properties

4.2.11 SIDESCAN

Here you can select which sidescan sonar you want to interface into NaviScan – it can either be standalone SideScan or MBE SS systems (eg R2Sonic Snippets).

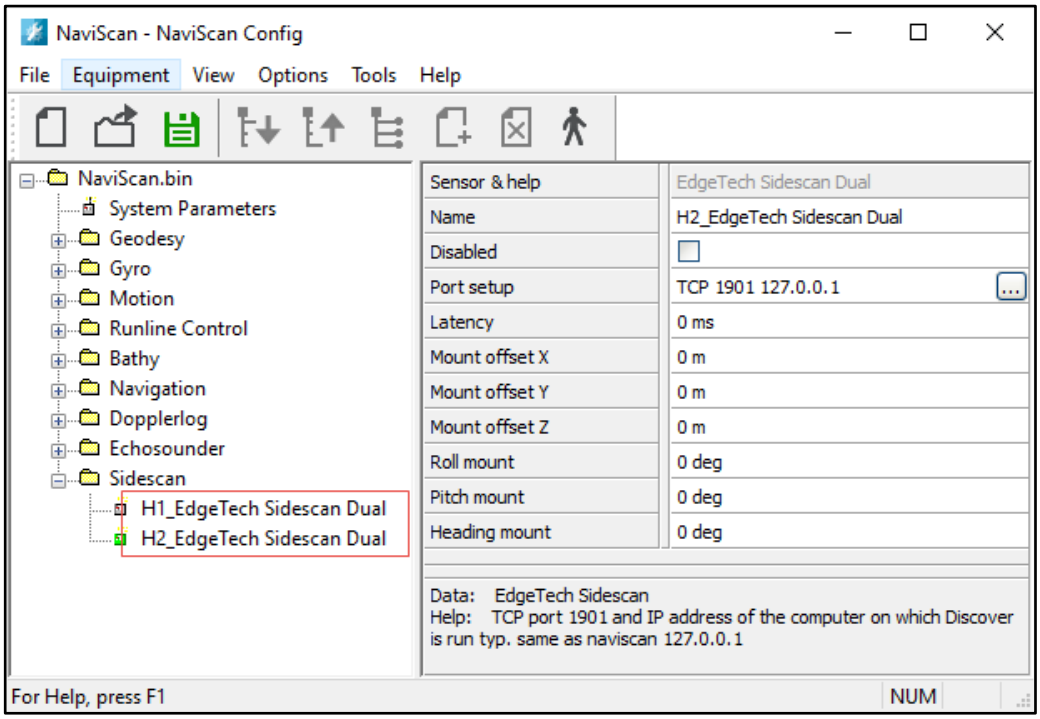


Figure 28 Edgetech sidescan sonar

4.2.12 PIPE TRACKER

Specify the pipe tracker system you are using as shown in the dialog below:

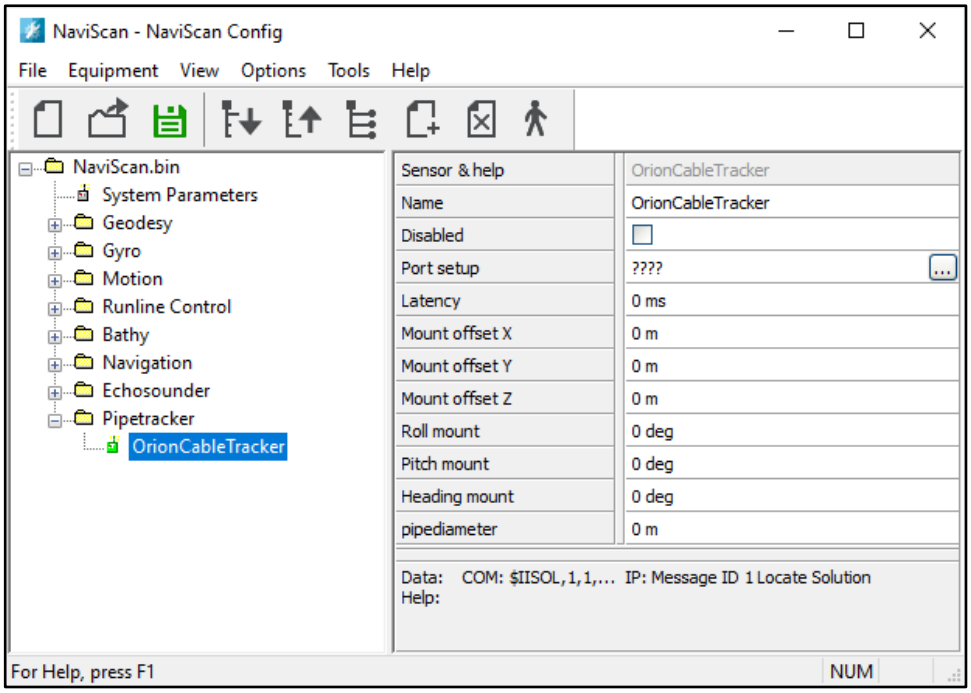


Figure 29 Sensor interfacing: Pipe tracker and its sensor properties

The pipe diameter can be defined in the sensor's properties. This value is used for calculating the correct depth.

Pre-defined pipe tracker sensors: DOF Subsea, Innovatum, Optimal Ranging (Innovatum), Reson, Teledyne, Tinsley.

4.2.13 LASERSCANNER

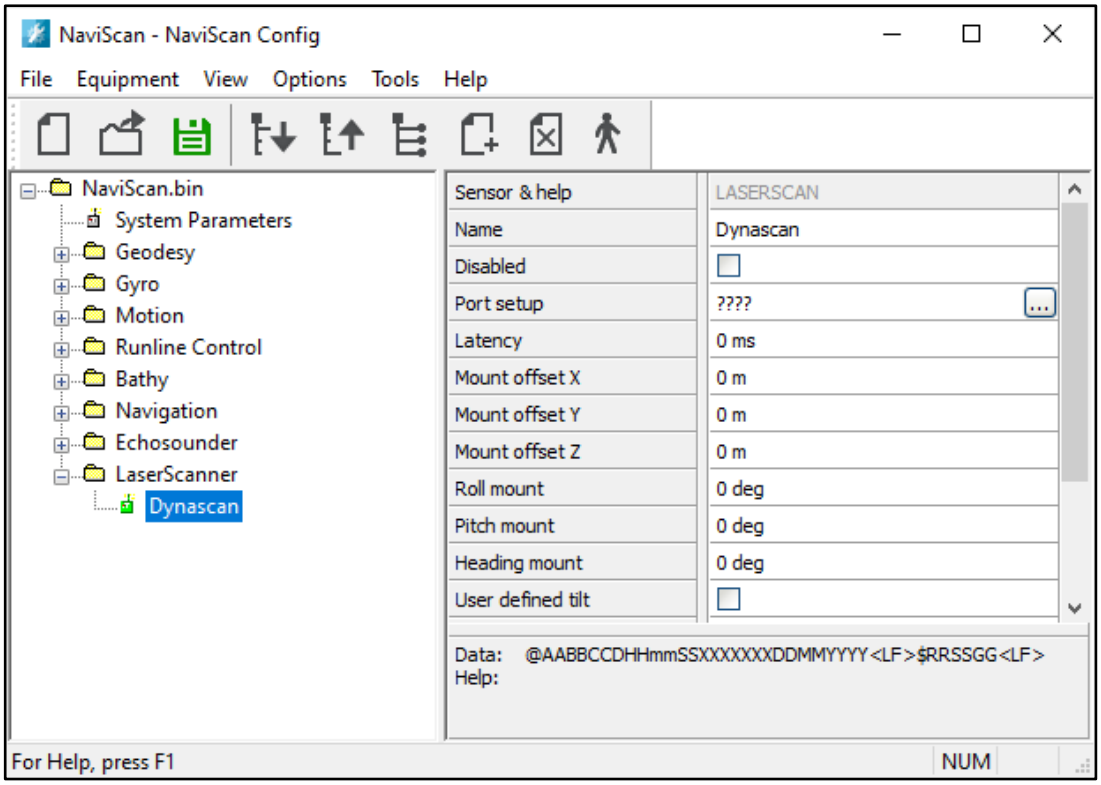


Figure 30 Laser scanner and its sensor properties

4.2.14 Free data format

The Free data format accepts data in any fixed position ASCII string with a known termination.

The sensor type **Free setup** is used, if the sensor you wish to interface is not listed and has an ASCII output.

Free Format	The Free setup shows how the input format is defined for the predefined sensors. Only when using the free format these fields can be edited; offsets for the relevant data are specified here. NOTE: The offset is the number of characters skipped before the actual parameter is read.
String Layout	The String Layout shows an ASCII string generated on basis of format defined at Free setup. When the free format is selected the String Layout isn't generated before clicking the Generate String button.
Termination character	For the free format, you most enter the binary code for string termination, eg 10 for <LF> or 13 for <CR>

To avoid any confusion, a few examples are shown below, describing how to set offsets.

As the procedure is the same for each sensor/free input, one example motion sensor is described.

Example **Free motion sensor**:

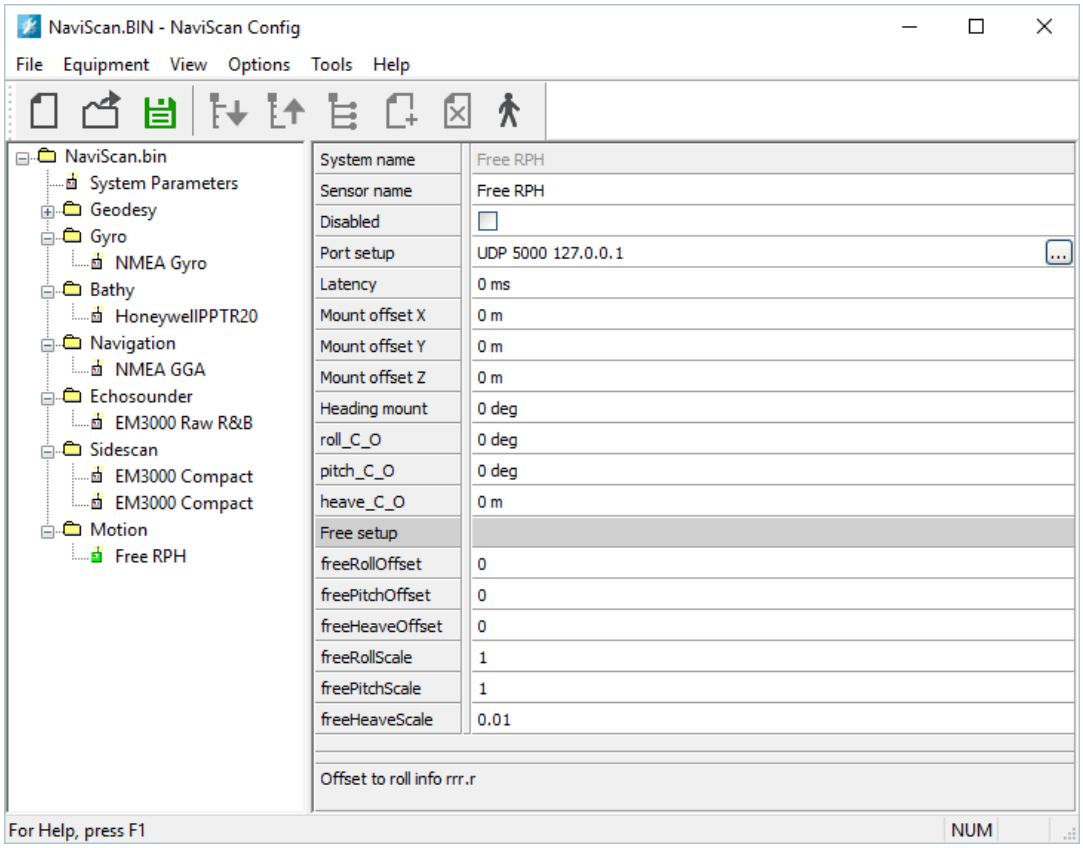


Figure 31 Roll and Pitch and Heave setup

Our ASCII data could look like this. (All examples below would work with values set above)

```
                    1         2       3
0123456789012345678901234567890
xx-5.345xx+3.234xxxxxxx0.321xxxxxx…
xx1.654xxx-2.234421xxxx-1.645xxxxx…
xx1.1xxxxx4.21xxxxxxxxx1.23xxxxxxx…
```

As you can see, you only have to set the offset value to the start of where you expect to get the data including the sign.

IMPORTANT:

As this is a generic driver, there is one thing, which is very important.

Just because the **Serial string** in the dialog shows "***rrr.rr…..",** doesn't it mean, that it will only read 6 characters..

It reads the floating point number, which comes directly after the offset.

Example (roll offset = 2):

Ad-2.42yuiweryu… -> Roll = -2.42
122.474272yeuqwi -> Roll = 2.474272
09+12.6a5252 -> Roll = 12.6

4.2.15 Special instruments

Some instruments require special setting that aren't handled in the general instrument setup:

RDI DVL	NaviScan expects that the DVL is mounted correctly (or compensated as such inside the DVL). We have seen some scenarios where it has been rotated 180 degree. To get an optimal behaviour you may compensate for this (negating the readings)
Applanix POSMV	This relates purely to the network interface towards Applanix POSMV • Attitude: NaviScan reads as default attitude (gyro and roll/pitch/heave) from the 102 packet. You may change this to use packet 103 instead. • Position: The position is as default taken from the special position packet (1). You may alternatively use the attitude packet (102(103)
Digiquartz	NaviScan expects that the Digiquartz data is given in PSI. If this isn't the case then you may enter a scale divider here. Pressure PSI = Pressure Observed / Divider Eg if data is in MBAR then enter 68.94757
Heave	The selected unit has been observed sending heave in both upwards and downwards reference (positive up or positive down) coordinate system. NaviScan has no way of detecting this - so you may have to tell NaviScan to change sign of the heave.

	This has so far been observed for the following units Seapath Ixsea GC2 Ixsea RB To evaluate the heave please note that NaviScan expects the heave to be positive below datum – eg a value added to the measured depth

4.3 View

Facility to toggle the Status Bar on and off.

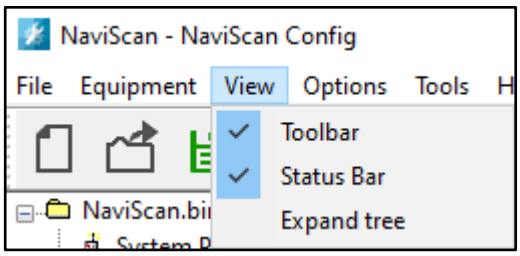


Figure 32 View options

- Toolbar
- Status Bar
- Expand tree

4.4 Options

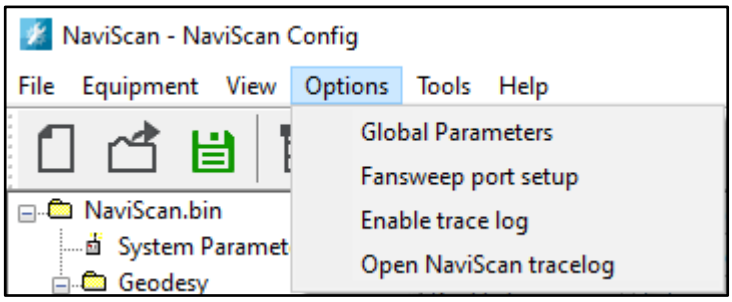


Figure 33 Options Global parameter, Fansweep port setup, Enable trace log, Open NaviScan tracelog

4.4.1 Global Parameters

Selecting this menu point, opens a dialog where you can setup different online settings.

It is common to all settings that they have influence on the online behavior and it will not be effective until NaviScan Online has been restarted.

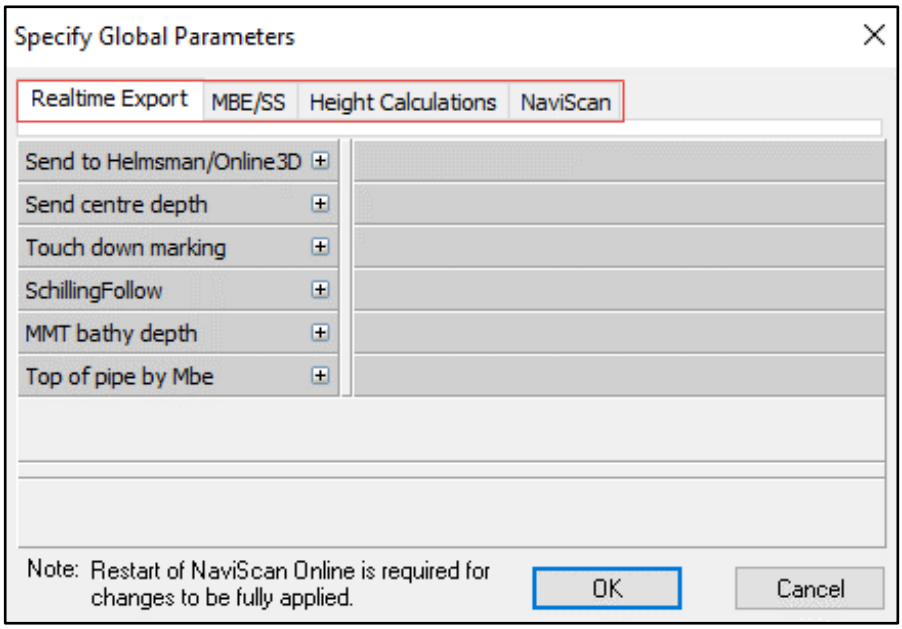
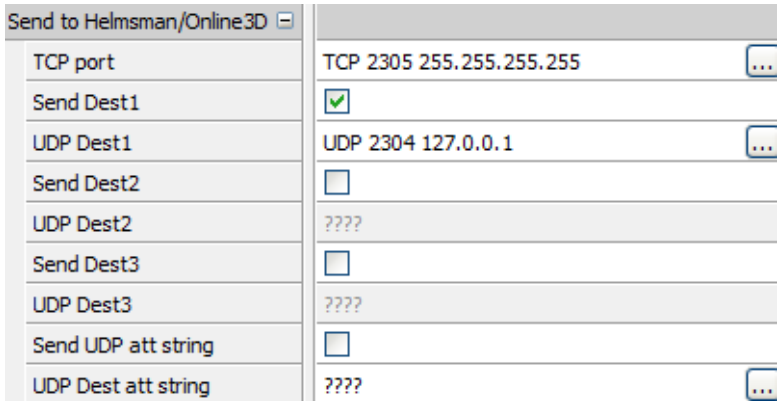
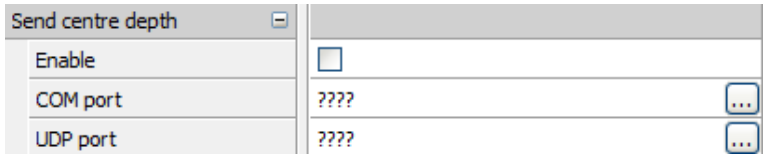
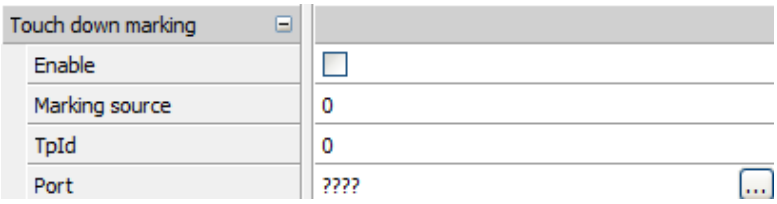
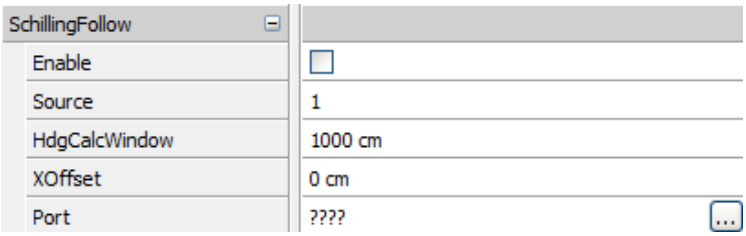
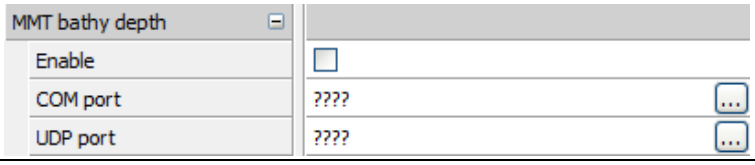
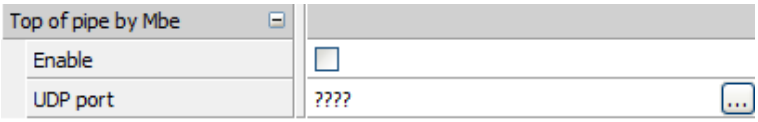


Figure 34 Specify Global Parameters

- **Real-time Export**

  NaviScan offers the user besides data recording various outputs to data QC in a.o. NaviPac

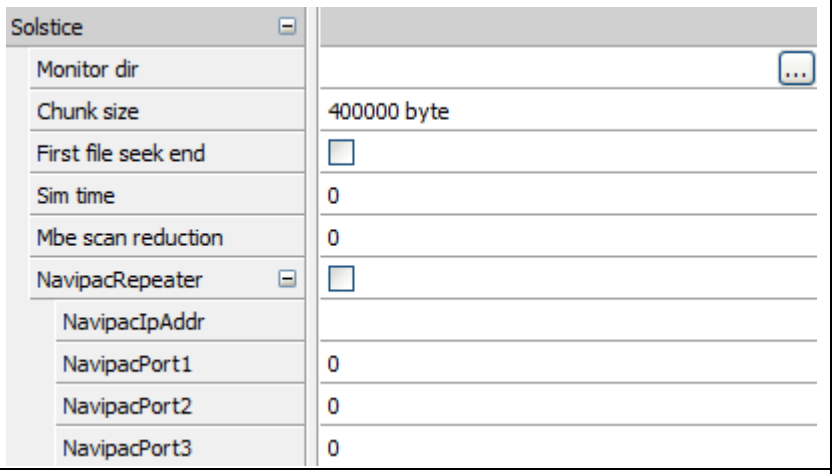
Option	Description
Send to Helmsman/Online3D	 NaviScan can generate a digital terrain model similar to what's displayed inside NaviScan online and send this out on UDP/IP port. This is compatible with the data expected in NaviPac's Helmsman's Display. The administrator must specify a few parameters on this issue Enable/Disable DTM Turn on/off the DTM calculation and export. Do not enable unless you are using it! Port settings Identify the UDP/IP (network) address and port for the export. Default is port 2310 Enable/Disable SS DTM Turn on/off the sidescan DTM calculation and export. Do not enable unless you are using it! Port settings Identify the UDP/IP (network) address and port for the export. Default is port 2311 Recording only Shall NaviScan send the DTM out only when recording data (selected) or all the time? Cell Size The NaviScan DTM is based on quadratic cells – and the operator must predefine the size of the cells. Cell sizes must be determined carefully. (normally between 100 – 500 cm), normally less or equal to the final cell size used in Helmsman's Display
Send centre depth	 NaviScan calculates and outputs the centre depth (the most nadir depth) to external use in eg NaviPac. The format of the output is in NMEA style.

	`$NBDBT,,,<Depth>,M,,<cr><lf>`, where <Depth> is the centre-beam depth. Example: `$NBDBT,,,23.43,M,,<cr><lf>` When the tick box is ticked, the message will be broadcasted on internal UDP/IP port (127.0.0.1 port 5999). You may furthermore select a serial port for the output.
Touch down marking	 NaviScan offers a utility where you may mark targets (eg Pipe touch-down) on the online DTM view and export the position to an external client. In this section you enable this function and define the output settings. Please refer to the NaviScan manual **NaviScanTouchDownMonitor.pdf** for details
SchillingFollow	
MMT bathy depth	
Top of pipe by MBE	 Multibeam derived top of pipe send out for every scan of MBE 1

- **MBE/SS**

In this section you can control settings for various MBE sonar's

SeaBat 8000	• Output UTC telegram NaviScan will, if selected, send a special UTC telegram out on serial port. This will enable the Reson system to synchronise with the NaviScan pc time. This or similar is required if the Reson 8xxx is interfaced via LAN.
Seabat 7k flexmode	• Flexmode control src: Select seabat7k sonar flexmode beam steering control source to dynamically follow an object. • flexmode update reduction Reduce the update rate of flexmode steering commands by this number, this way downsampling the controlling source input. • flexmode offset angle: Additional offset to be added to all flexmode commands, positive at steering to starboard, in static mode it give the offset from nadir, in dynamic adds a fixed angle to the calculated steering angle.
Norbit Seabat 7k	Norbit 7k always subscribe

Edgetech4600	MaximumSwath in m. MaximumDepth in m. MinimumDepth in m Use NaviPac Repeater npIpAddr 127.0.0.1 npPosPort 2320 npRphPort 2321
Klein	Klein_calfile Klein_timestamplocaltime Klein_5900_wingIP Klein_5900:wingPort Klein:soundvelcm
Solstice	
Cathx	Select the directory to be monitored for pcp file data transfer.

- **Height Calculations tab**
  Define the height reference for vessel mounted systems

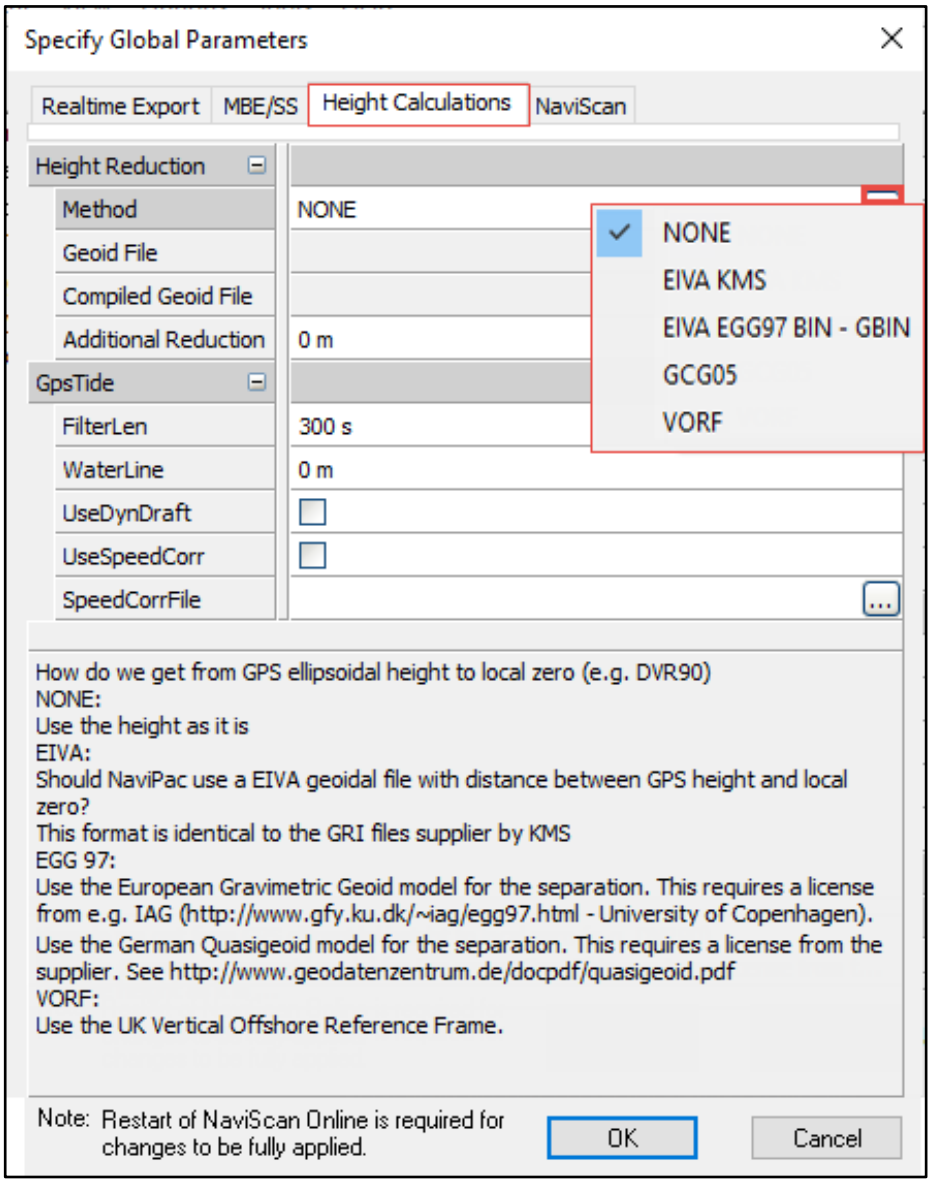
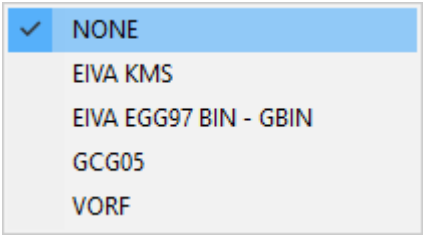


Figure 35 Global Parameters: Height Calculations

Height Reductions	These Height Reduction settings only apply to the instruments: **GPSTideGGA** and **GPSTidePosmv**. The instrument **GpsTideNP** is not affected by these settings and will be logged exactly as received from NaviPac.

Method	If NaviScan is used together with a GPS based position system (eg NMEA or POSMV input) then you may control the height reference and thus change from WGS84 to local zero **Method**: How do we get from GPS ellipsoidal height to local zero (eg DVR90)  • **NONE** Use the height as it is • **EIVA KMS** Should NaviScan use a EIVA geoidal file with distance between GPS height and local zero? This format is identical to the GRI files supplied by KMS – sample file found at \eiva\navipac\setup if using NaviPac • **EIVA EGG97 BIN – GBIN** Use the European Gravimetric Geoid model for the separation. This requires a license from eg IAG (http://www.gfy.ku.dk/~iag/egg97.html - University of Copenhagen). • **GCG05** Use the German Quasigeoid model for the separation. This requires a license from the supplier. See http://www.geodatenzentrum.de/docpdf/quasigeoid.pdf . **VORF**: Use the UK Vertical Offshore Reference Frame
Geoid File	Reference to the geoid file with the height reduction – only valid for EIVA, GCG05, VORF or EGG97 files.
Compiled Geoid File	NaviScan uses a compiled binary version of the source file. It's saved at same location with a .BIN extension.
Additional Reduction	Can be used as sole reduction (from WGS84 to normal zero) or as an extra reduction so the WGS height is compensated both for model based and fixed reduction.

GpsTide	These GpsTide correction settings only apply to the instruments: **GPSTideGGA** and **GPSTidePosmv**. The instrument **GpsTideNP** is not affected by these settings, and will be logged exactly as received from NaviPac.
FilterLen	The number of seconds in an average window, the longer the smoother gpstide (see IgnoreMotion comment), timestamp of gpstide is deskewed to the half of averaging window lenght
WaterLine	WaterLine is the z distance from the ships ref point to the projection of the ref point on the sea surface plane, under normal working load, z positive up, only used when no present dyndraft is enabled, and only for GpsTideGPS and GpsTidePosmv calc.
UseDynDraft	Use the first present dynamic draft type bathy sensor in setup, instead of fixed waterline (if a dyndraft is not present, waterline is still used), only used for GpsTideGPS and GpsTidePosmv calc
UseSpeedCorr	
SpeedCorrFile	

- **NaviScan**

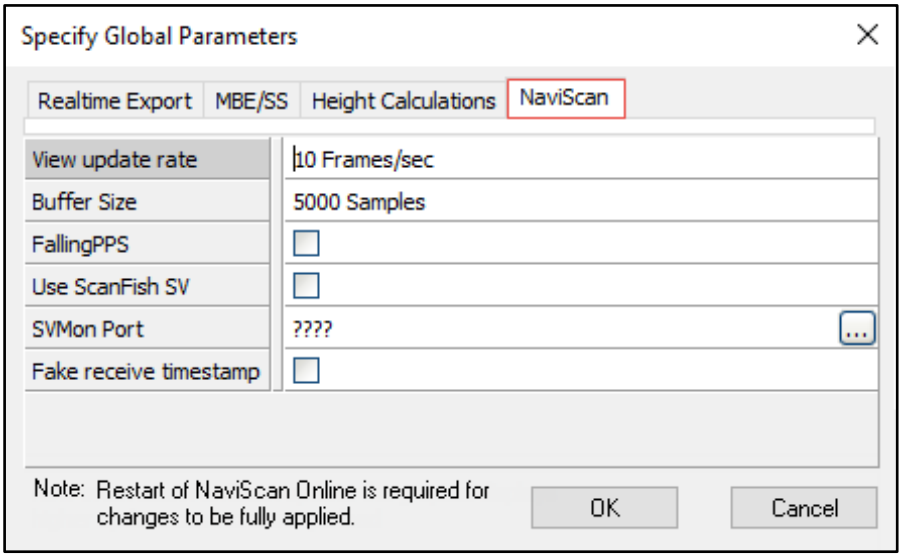


Figure 36 Specify Global Parameters: NaviScan tab

This section defines various settings related to NaviScan performance. Very rarely changed!

Table 1 Global parameter and the NaviScan tab

View update rate	How often must NaviScan update the graphical views. Use with care
Buffer Size	How many samples do you want to be displayed in the time-based windows in the NaviScan Online program. Note: The bigger the amounts of samples, the more CPU load!!
FallingPPS	For GPS time with 2 trig per cycle, trig on falling edge, instead of default rising edge.
Use Scanfish SV	When using NaviScan together with a ScanFish you may get online updated sound velocity profiles (via the SVMon program) and apply that to the data for improved online QC
SVMon Port	
Fake receive timestamp	

4.4.2 Fansweep port setup

Selecting this item opens the configuration file (StnAtlas.ini) in notepad, where the individual ports for the Atlas fansweep MBE (bathymetry) and the sidescan can be seen. The IP address is set in the normal way using interface setup but can be overwritten for a specific item in the **StnAtlas.ini** file

StnAtlas.ini - Notepad

File Edit Format View Help

```
[fansweep]

; optional Ipaddres added this way:  fs20_xxxx=port 192.168.1.4
fs20_lot=3100
fs20_depth_1=3111
fs20_depth_2=3112
fs20_depth_3=3113
fs20_depth_4=3114
fs20_depth_5=3115
fs20_depth_6=3116
fs20_depth_7=3117
fs20_depth_8=3118
fs20_sidescan_1=3125
fs20_sidescan_2=3126
fs20_sidescan_3=3127
fs20_sidescan_4=3128
fs20_angle_1=3141
fs20_angle_2=3142
fs20_angle_3=3143
fs20_angle_4=3144
fs20_angle_5=3145
fs20_tpe=3159
fs20_offset=25000


[svp2fansweep]
XDR Port=3070
z-keel=0.050


[hydromap]
; 8:"MD3",9:"DS3",10:"PS3",11:"PS3_35"
hmsystemid=9
hmmbechannel=es_targets_port
hmmotionchannel=fs20_motion_1
hmasddatatype=8
```

Figure 37 Atlas fansweep

The **Fansweep Port setup** option opens a file, **StnAtlas.ini**, where the individual ports for a Fansweep MBE with sidescan option can be defined.
The instrument is configured in the normal way, via the equipment/interface configuration. Once this is accomplished, the StnAtlas.ini file must be configured along the following guidelines:
There are 6 types or packages, for which individual ports must be defined:
a. „fs20_lot“: Standard parameters for echo sounder
b. fs20_depth_1 to _8: Bathymetry packages containing depth

c. fs20_angle_1 to _5: Bathymetry packages containing angle
d. fs20_sidescan_1 to _4: Sidescan packages containing sidescan image data
e. fs20_tpe: Total Propagated Error telegram
f. fs20_offset:a port offset will be added to all the port numbers if this line is present anywhere in the StnAtlas.ini file eg fs20_offset=25000.
To change the IP address for the TPE telegram, which seems to be required for fs20 (or to change the IP address for a specific item in general) just write the IP address behind the item, separated by a space: fs20_tpe=3159 192.168.1.89

For a more detailed description of these ports and packages, refer to the general Atlas Fansweep documentation.

4.4.3 Enable trace log

Activating this item will toggle the **Error trace log** on and off. The traces can be used for debugging.
NaviScan Online will hereafter generate dump files in the NaviScan\bin folder which could help EIVA tracking down problems. Please send these files to support@eiva.com

4.5 Tools

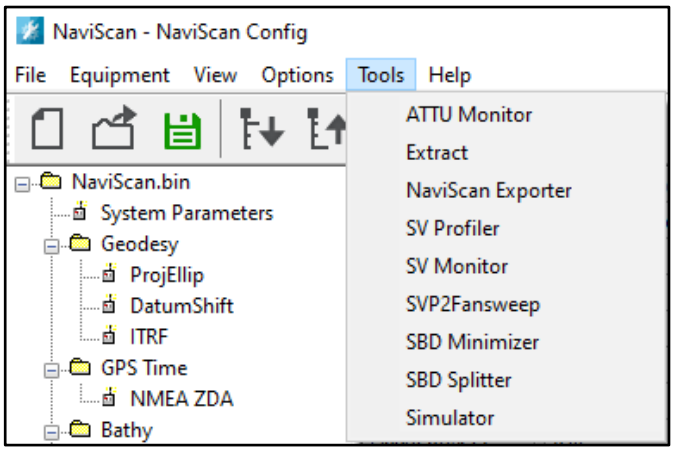
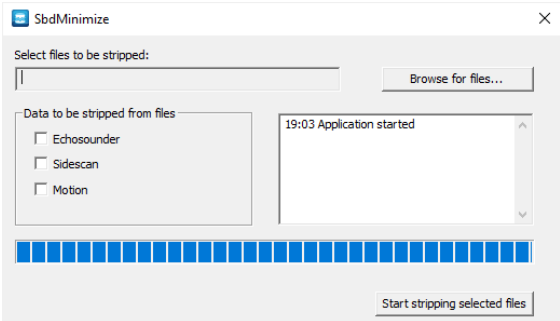


Figure 38 NaviScan Configuration: Tools

ATTU Monitor	Use for ATTU Time Synchronization program (AttuMon.exe). It was created for the purposes of monitoring the ATTUs and synchronization the system time of the computer with one of them.

Extract	To extract information to MS Excel readable files for quality control
NaviScan Exporter	To export to fixed ASCII and binary XYZ format, to convert the files to XTF format
SV Profiler	To read/show sound velocity profiles
SV Monitor	To monitor sound velocity read from ScanFish software
SVP2Fansweep	For Hydrosweep inbound server, these setup items are used (here with default settings): hmsystemid=9 hmasddatatype=8 hmmbechannel=es_targets_port hmmotionchannel=fs20_motion_1 For the SVP to Fansweep program these items are used: XDR Port=3070 z-keel=0.050
SBD Minimizer	
SBD Splitter	To split an SBD file, by kp or size
Simulator	To simulate different sensors

4.6 Help

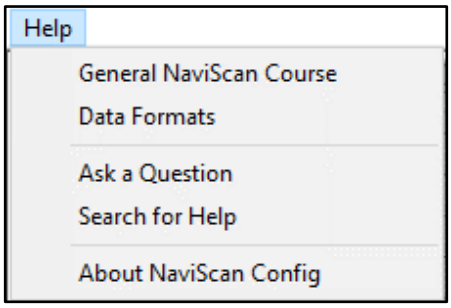


Figure 39 Help

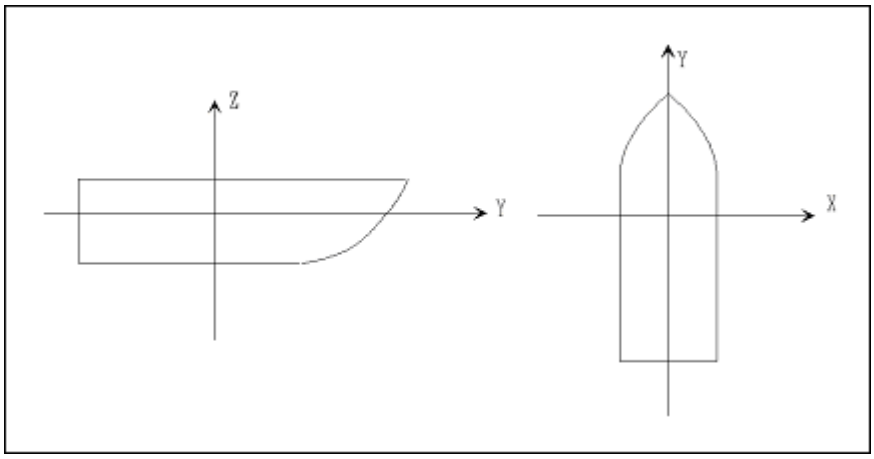
- **General NaviScan Course**
  Shows some general slides used in basic NaviScan Training Course (not updated)
- **Data Formats**
  NaviScan Reference Manual with Sensor formats (not updated)
- **Ask a Question**
  Opens your default mail program and to pass us an email to **support@eiva.com**
- **Search for Help**
  Opens your default web browser and searches the web
- **About NaviScan Config**
  Opens the **About** box with useful information about **Product Version**, and **Release Date**. This information should be given to EIVA Support when issues and observations are reported.
  Licence Info
  Find the **PC Code** here.

5 Definitions

All depending on the type of sensor offset and mount angles can be set. Below is described how these are defined.

5.1 Offsets

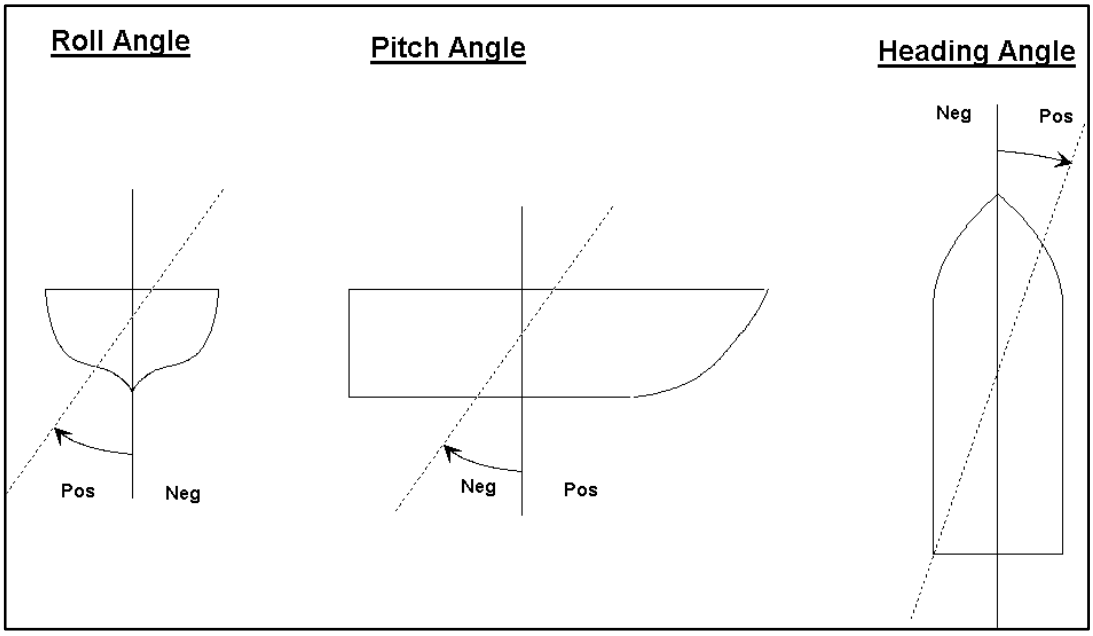
The offset is defined as the (X,Y,Z) distance [m], from the primary position/navigation sensor, and to the sensor, where X is positive starboard, Y is positive front and Z is positive up, as shown in figure below.



5.2 Mount angles

The mount angle defines the exact alignment of the sensor.

The mounting is specified by roll angle, pitch angle and heading angle; the roll angle is measured positive starboard down, the pitch angle positive bow up and the heading angle zero front and positive clockwise as shown in the figure below:



6 Version descriptions

Version	Date	Author	Description
9.0	21.10.2016	ANS	Transferred to new template
9.4	27.05.2020	ANS	Reviewed by ANS and ALF